NearSens - project documentation

Contactless distance meter 100-500 mm

Field	Value
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Document version	1.0
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Schematic version	v1.0.0
PCB version	v1.2.0
Firmware version	v1.0.0

1. Purpose and scope

This document describes the construction and operation of the NearSens prototype. It covers the electronics, firmware, calibration, PCB production files, bill of materials and mechanical models used to build the device.

NearSens measures the distance to an object with an ultrasonic time-of-flight method over the practical 100-500 mm range. The result is shown on a four-digit LCD. The circuit is powered from an external 5 V DC source.

2. Prototype status

Area	Status
Schematic	v1.0.0, SVG file and EasyEDA JSON source
PCB	v1.2.0, EasyEDA JSON source, board outline DXF and Gerber
Firmware	v1.0.0, Arduino/C++ source code
Enclosure	STL model v1.0.0 and bracket
BOM	CSV
Revisions	earlier schematic, PCB, firmware and enclosure files in the revisions folder

3. Design assumptions

Area	Selected solution
Measurement method	ultrasonic echo time-of-flight measurement
Practical range	100-500 mm
Power supply	5 V DC through connector PS1
Result display	four-digit LCD
Controls	two buttons: TS3 and TS4
Calibration	100, 200, 300, 400 and 500 mm points
Settings memory	microcontroller EEPROM
Microcontroller clock	5 MHz
CAD documentation	schematic and PCB in EasyEDA JSON
PCB production	Gerber files v1.2.0
Mechanical parts	STL models of the enclosure and bracket

4. Measurement principle

The microcontroller generates a short burst that drives the ultrasonic transmitter. The wave reflects from the object surface and returns to the receiver. The receiver signal passes through the analog path and is then sampled by the microcontroller ADC.

Distance is calculated from echo return time. The firmware analyses a series of samples and rejects results that do not meet stability conditions or appear outside the detection window. After calibration, the result is corrected using points stored in EEPROM.

5. Functional blocks

Block	Role in the device
5 V DC input	prototype power supply
ATtiny1626	measurement control, ADC, LCD, buttons and EEPROM
ST232CDR	ultrasonic transmitter drive
40 kHz transmitter	ultrasonic burst emission
40 kHz receiver	echo reception from the object
LM324	analog receiver path
MC14543BDG	LCD segment driving
LCD DE117	result and diagnostic code display
TS3 and TS4	measurement, calibration and mode handling

6. Transmit path

The transmit path uses the ATtiny1626, the ST232CDR and a 40 kHz ultrasonic transducer. The microcontroller drives the T1IN and T2IN inputs of the ST232CDR. This drive method gives stronger transducer excitation than direct driving from a microcontroller port.

The firmware sends a 16-cycle burst. The drive half-period is 12 us, which gives excitation near 41.7 kHz. The detection window is limited to the range needed for 100-500 mm operation, so late reflections unrelated to the measurement are not analysed.

7. Receive path

The receive path consists of the ultrasonic receiver, LM324 stages and the microcontroller ADC input. The receiver signal is analog, so it is not treated as a single logic state. The firmware searches for the echo in a series of samples and checks amplitude and result stability.

In the prototype, R4 has a value of 220 ohm. This part is listed separately in the BOM.

8. Microcontroller and firmware

The firmware is located in firmware/NearSens_1.0.0. The main sketch file is NearSens_1.0.0.ino. The remaining files split the code into button handling, calibration, display, hardware setup, measurement, device state and helper functions.

The code is prepared for Arduino IDE with the megaTinyCore package. The package contains source files. Build configuration and compiled output files are not recreated in this documentation.

The J_PROG pad exposes the ATtiny1626 UPDI line. In the prototype, the series resistor for this line is not soldered on the board, so programming requires an external 4.7 kOhm resistor in the UPDI wire.

File	Code area
NearSens.h	pin definitions, constants, states and function declarations
Buttons.cpp	TS3, TS4, RAW mode, 75 mm correction and calibration entry handling
Calibration.cpp	100-500 mm calibration and EEPROM
Display.cpp	LCD handling and display codes
Hardware.cpp	pin setup, ADC setup, TX path power control and sleep handling
Measurement.cpp	ultrasonic measurement and echo qualification
State.cpp	firmware state variables
Utils.cpp	helper functions

9. User interface

The device can be started and checked without a computer. A 5 V DC power supply is enough for a basic test.

Element	Function
PS1	5 V DC power input
TS3	single measurement and calibration point storage
TS4	75 mm correction toggle and calibration point selection
TS3 + TS4	enter calibration after holding for about 5 s
5 TS3 clicks	toggle RAW mode
LCD	measurement result, work modes and error codes

10. Calibration

Calibration uses the 100, 200, 300, 400 and 500 mm points. For each point, the device performs a series of measurements and stores the accepted value in EEPROM. Later measurements are corrected using the stored points.

Calibration quality depends on the test setup geometry, object surface, transducer alignment and ambient conditions. The object should be flat, hard and perpendicular to the transducers.

11. Temperature effect

The speed of sound in air depends on temperature. The prototype does not measure ambient temperature, so the firmware does not compensate for this effect. This does not block operation in the selected range, but it limits claims of accuracy independent of test conditions.

A possible improvement would be adding a temperature sensor and using its reading when converting time of flight to distance.

12. Display codes

Code	Meaning
8888	device startup and ready for measurement
1111	calibration point stored correctly
7001	RAW enabled
7000	RAW disabled
0 _ 75	75 mm reference correction disabled
1 _ 75	75 mm reference correction enabled
4000	general no-result code
4001	no stable echo
4002	echo too early, crosstalk or transmitter blanking
4003	echo outside technical range
4004	calibration error
4005	no valid calibration data
4006	calibration sample spread too large
4007	ADC/RX receiver path error

13. PCB and production files

Hardware documentation is located in the hardware directory. EasyEDA JSON files are the CAD sources for the schematic and PCB. Gerber files are intended for PCB fabrication.

Folder	Contents
schematic	schematic SVG and EasyEDA JSON source
pcb	EasyEDA PCB source, DXF outline and layer previews
gerber	PCB production files v1.2.0
enclosure	STL models of the enclosure and bracket

14. Mechanical documentation

The enclosure and bracket are mechanical parts of the prototype. Their cost is not included in the electronic parts and PCB BOM.

File	Description
NearSens_enclosure_v1.0.0.stl	main enclosure model
NearSens_bracket_v1.0.0.stl	bracket used during assembly

15. BOM

The BOM is in the bom folder in CSV format. It includes electronic parts and the PCB. Prices are given in USD.

The total value of the BOM items is 11.7949 USD. The enclosure is not included in this value. R4 is listed in the BOM as a part from local stock.

16. Project revisions

The revisions folder contains earlier versions of the schematic, PCB, firmware and enclosure. This material shows the development process. Current production files are located in the main docs, hardware, firmware and bom folders.

17. Main file contents

File or folder	Meaning
docs/NearSens_user_guide.pdf	startup, use and calibration
docs/NearSens_project_documentation.pdf	project description
hardware/schematic	schematic and CAD source
hardware/pcb	PCB sources, outline and previews
hardware/gerber	PCB production files
hardware/enclosure	STL models
firmware/NearSens_1.0.0	source code
bom	bill of materials
revisions	earlier mechanical, firmware, schematic and PCB revisions